

	the sphere.]
9(a)(i)	The principle of superposition states that when two or more travelling waves of the same type meet at a point in space, the resultant displacement at that point is the vector sum of the displacements due to each of the waves at that point.
9(a)(ii)	Coherent light waves mean that the waves have a constant phase difference.
9(b)(i)	Path difference is $\frac{3}{2}\lambda$
9(b)(ii)	The path difference between the light from two slits when the light reaches R is $\frac{3}{2}\lambda$. The two coherent waves arrive antiphase with a phase difference is 3π radian. Hence destructive interference occurs at point R and the resultant displacement is minimum.
9(b)(iii)	Using $x = \frac{\lambda D}{a}$ $2.9 \times 10^{-3} = \frac{\lambda \times 2.70}{0.60 \times 10^{-3}}$ $\lambda = 6.44 \times 10^{-7} \text{ m} = 6.4 \times 10^{-7} \text{ m}$
9(b)(iv)	Light passing through a single slit spread out due to diffraction, forming its own pattern. The intensity of this pattern is brightest at the centre and decreases towards the edges. Since the slits separation is close to each other, the diffraction pattern due to both slits overlap. The central maximum has the highest intensity because that is the position where both double slit constructive interference and single slit diffraction peak occur. At other observable maximum, constructive interference still occur but their brightness is reduced as they lie where the single slit diffraction intensity is lower.

Qn	Solution
9(c)	Intensity $I = kA^2$ ----- (1) $\frac{I}{2} = k(A')^2 \text{ --- (2)}$ $\frac{(2)}{(1)} \quad A' = \frac{1}{\sqrt{2}} A$ Resultant intensity at P $I_R = k(A + A')^2 = k\left(A + \frac{1}{\sqrt{2}}A\right)^2$ $= k\left[\left(1 + \frac{1}{\sqrt{2}}\right)A\right]^2 = 2.9kA^2 = 2.9I$
9(d)(i)	Using $d \sin \theta = n\lambda$ $\tan \theta = \frac{1.12}{2.70} \Rightarrow \theta = 22.53^\circ$ $1.67 \times 10^{-6} \sin 22.53^\circ = 1 \times \lambda$ $\Rightarrow \lambda = 6.40 \times 10^{-7} \text{ m}$
9(d)(ii)	The bright fringes formed using a diffraction grating are sharper and narrower. The fringes are also of high intensity and larger separation (1.12 m compared to 2.9 mm) than those produced by double slits. These sharp fringes are easier to locate accurately on the screen, reducing uncertainty in position measurement and results in smaller uncertainty when calculating the wavelength.